

Module 7

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Module 7

Advanced Inheritance

In this module we learn more about inheritance, starting with the idea of abstract data types (ADTs). An ADT is a class that is intended as the base of a derivation hierarchy. In fact, you can't even create an object of an ADT type, because the ADT's type definition is "incomplete." Nevertheless, ADTs play an important role in object-oriented design.

We also visit the topic of multiple inheritance, which means a derived object inherits properties of two or more base objects. Unfortunately, multiple inheritance often brings with it as many problems as it solves. It can be profitably used in certain situations, some of which we will encounter, but use it sparingly, and generally only as a last resort.

Finally, we encounter private derivations, which prevent a single type of access. The prohibition is, however, quite important in maintaining the integrity of derived objects.

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Learning Objectives

- Abstract Data Types (ADTS)
- Multiple inheritance
- Private derivations

Lectures

Presentation

- [7.1: Comments on Reading Assignments for Module 6](#)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-7-dot-1>)
- [7.2: Abstract Data Types \(ADTs\)](#)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-7-dot-2>)
- [7.3: Private Derivation](#)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-7-dot-3>)
- [7.4: Graphs](#) (<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-7-dot-4>)
- [7.5: Practical Example of Private Derivation](#)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-7-dot-5>)
- [7.6: Multiple Inheritance](#)
(<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-7-dot-6>)

Reading

The reading for this Module is C review. Depending on your C background, you may be able to move through this material quickly.

Required Readings

- *C++ Primer Plus*, Chapter 14

Recommended Readings

- *Thinking in C++ Volume One*, pp. 531–570, 647–679

Discussion

[Module 7 Discussion](#)

(https://onlinelearning.berkeley.edu/courses/2291660/discussion_topics/14351908)

Assignment

[Module 7: Short Answer 7](#)

(<https://onlinelearning.berkeley.edu/courses/2291660/assignments/33623162>)

[Module 7: Programing Assignment 7](#)

(<https://onlinelearning.berkeley.edu/courses/2291660/assignments/33623160>)

Source Code

In each module of this course, you will be required to submit a programming assignment. For each module, I have prepared sample source code for your reference. Please take a few minutes to download and install all of these programming examples. The files have been compressed into one zip. You will need a zip utility to decompress the file.

[sources.zip](#)

(<https://onlinelearning.berkeley.edu/courses/2291660/files/313890817/download?wrap=1>)

